

Module 4: Supervised Learning
Submodule: 4.2 Logistic Regression
Lecture: Logistic Regression and Classification

XCS229 Machine Learning

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I. Introduction to Classification

We will discuss the concept of classification, specifically focusing on how we represent data points and their corresponding labels.

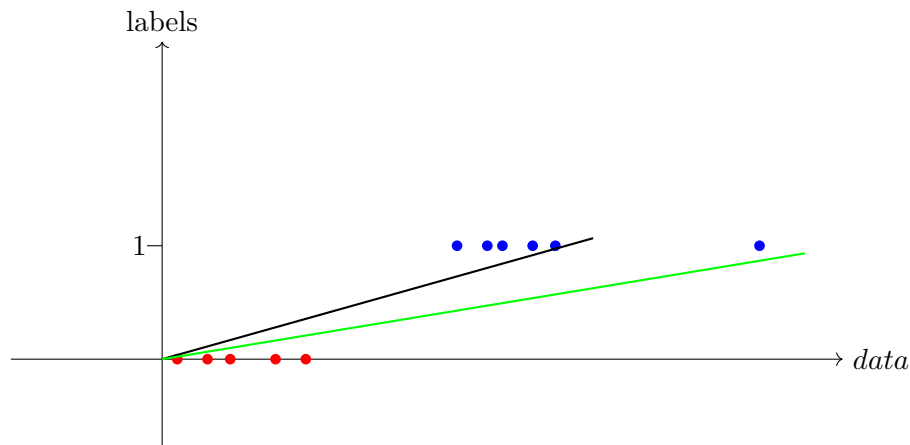
II. Data Representation

Given $\{(x^{(1)}, y^{(1)}) \text{ for } i = 1, \dots, n\}$.

- $x^{(i)} \in \mathbb{R}^{d+1}$, where there are d features plus a bias term
- $y^{(i)} \in \{0, 1\}$, where:
 - $y_i = 0$: negative class (e.g., not a cat or no tumor).
 - $y_i = 1$: positive class (e.g., is a cat or there is a tumor).

III. The Problem with Linear Regression

To visualize the inadequacy of linear regression for classification tasks, consider a one-dimensional feature space.



If we plot the data, we might try to fit a line to separate the classes based on the threshold (e.g., at $y = 0.5$). The issue arises when outliers are present, causing the line (e.g., the green line) to stretch infinitely.

IV. Logistic function

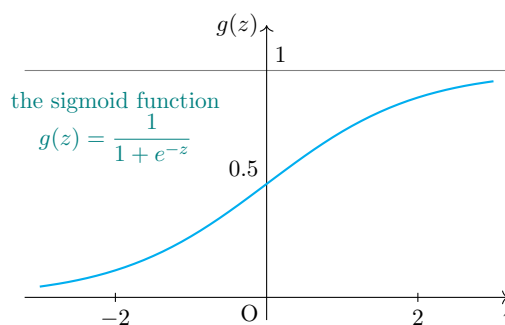
We want our hypothesis h_θ to predict a value between 0 and 1 ($h_\theta(x) \in [0, 1]$) instead of shooting out to infinity. To fix this, let's change the form for our hypotheses $h_\theta(x)$. We will choose

$$h_\theta(x) = g(\theta^T x) = \frac{1}{1 + e^{-\theta^T x}}$$

where

$$g(z) = \frac{1}{1 + e^{-z}} \quad (\text{link function})$$

is called the **logistic function** or the **sigmoid function**. Here is a plot showing $g(z)$:



Notice that $g(z)$ tends towards 1 as $z \rightarrow \infty$ and $g(z)$ tends towards 0 as $z \rightarrow -\infty$. Moreover, $g(z)$, and hence also $h(x)$, is always bounded between 0 and 1.

We interpret $h_\theta(x)$ as the probability that $y = 1$ given x :

$$P(y = 1 \mid x; \theta) = h_\theta(x)$$

And consequently,

$$P(y = 0 \mid x; \theta) = 1 - h_\theta(x)$$

Assuming that the n training examples were generated independently, we can write down the likelihood of the parameters as:

$$\begin{aligned} L(\theta) &= p(\vec{y} \mid \vec{X}; \theta) \\ &= \prod_{i=1}^n p(y^{(i)} \mid x^{(i)}; \theta) \\ &= \prod_{i=1}^n \left(h_\theta(x^{(i)}) \right)^{y^{(i)}} \left(1 - h_\theta(x^{(i)}) \right)^{1-y^{(i)}} \end{aligned}$$

The $y^{(i)}$ exponents are ‘encoding’ the “if” condition: if $y^{(i)} = 1$ then the first term is ‘active’, otherwise it’s the second term. Taking the logarithm of the likelihood simplifies calculations:

$$\begin{aligned} \ell(\theta) &= \log L(\theta) \\ &= \sum_{i=1}^n y^{(i)} \log h_\theta(x^{(i)}) + (1 - y^{(i)}) \log(1 - h_\theta(x^{(i)})) \end{aligned}$$

This expression represents the logistic loss, allowing us to proceed with optimization.

V. Maximizing the Likelihood

To maximize the likelihood, we can use a method analogous to the derivation we performed in the case of linear regression. We utilize gradient ascent for this purpose.

Written in vectorial notation, the updates are given by:

$$\theta^{(t+1)} := \theta^{(t)} + \alpha \nabla_\theta \ell(\theta^{(t)})$$

(Note that we use a positive sign in the update formula since we are maximizing rather than minimizing a function.)

Let’s start by working with just one training example (x, y) and take derivatives to arrive at the stochastic gradient ascent rule. We differentiate the log-likelihood function $\ell(\theta)$ with respect to the parameter θ_j :

$$\begin{aligned}
\frac{\partial}{\partial \theta_j} \ell(\theta) &= \left(y \frac{1}{g(\theta^T x)} - (1-y) \frac{1}{1-g(\theta^T x)} \right) \frac{\partial}{\partial \theta_j} g(\theta^T x) \\
&= \left(y \frac{1}{g(\theta^T x)} - (1-y) \frac{1}{1-g(\theta^T x)} \right) g(\theta^T x)(1-g(\theta^T x)) \frac{\partial}{\partial \theta_j} \theta^T x \\
&= (y(1-g(\theta^T x)) - (1-y)g(\theta^T x)) x_j \\
&= (y - h_\theta(x)) x_j
\end{aligned}$$

In the derivation above, we utilize the fact that the derivative of the logistic function $g(z)$ is given by $g'(z) = g(z)(1-g(z))$.

Thus, we obtain the stochastic gradient ascent rule:

$$\theta_j^{(t+1)} := \theta_j^{(t)} + \alpha \sum_{i=1}^n \left(y_j^{(i)} - h_\theta(x_j^{(i)}) \right) x_j^{(i)}$$

If we compare this update rule with the least mean squares (LMS) update rule, we observe that they appear identical. However, it is crucial to note that this is *not* the same algorithm because $h_\theta(x^{(i)})$ is now defined as a nonlinear function of $\theta^T x^{(i)}$.

Despite the differences in the underlying algorithms and learning problems, it is surprising that we arrive at the same form of the update rule.

The question arises: is this resemblance merely a coincidence, or is there a deeper reason underlying this phenomenon? We will address this intriguing inquiry when we explore generalized linear models (GLM).

1. Understanding Assumptions

- Assumptions are integral components that we introduce to the problem.
- They enable progress in machine learning.
- Currently, the assumptions made are relatively simple, primarily concerning functional forms.

2. Defining Probability Functions

- We assume that the probability of y equaling 1 adheres to a specific function.
- The chosen function is appealing as it:
 - Ranges between 0 and 1,
 - Is monotonically increasing,
 - Possesses certain desirable standard properties.

3. Complementary Probabilities

- The reason for using $1 - h_\theta$ is due to the binary nature of y :
- Since y can take only two values ($y \in \{0, 1\}$), the probabilities must sum to 1.

4. Simple Assumptions and Model Design

- The assumptions we make are straightforward linear models or linear models with a specific link function.
- Discussions can be made on why a particular link function, g , is chosen:
 - Potential exploration of different link functions based on structural properties.

5. Nature of Assumptions

- The assumptions made are not about proving truth; they are models that guide our analysis.
- By making such assumptions, we can derive significant results.
- The key question is how well these assumptions reflect the real world.

6. Independence in Assumptions

- The question of when independence matters is nuanced and cannot be definitively answered now.
- A discussion will be saved for later lessons where independence can be notably violated and affect outcomes significantly.

7. Modeling Realities

- Many variables in the real world are not independent, which can pose challenges.
- Understanding the limitations of assumptions takes time but can lead to effective modeling practices.

8. Probabilistic Framework

- The transition from full probability to products (due to independence) is critical in the current derivations where we break down data into distinct choices.

VI. Summary

In this lecture, we covered the following key takeaways:

- **Classification Concept:** We explored the fundamentals of classification, emphasizing how data points and their labels are represented.
- **Limitations of Linear Regression:** We identified the issues with using linear regression for classification tasks, particularly in the presence of outliers.
- **Logistic Function:** The logistic function was introduced, allowing us to predict probabilities within the range of $[0, 1]$, which is essential for binary classification.
- **Likelihood Function and Estimation:** We derived the likelihood function for our model, leading to the formulation of the log-likelihood and the concept of logistic loss for optimization.
- **Gradient Ascent:** We explained how to maximize the likelihood using gradient ascent and how this method resembles the update rule used in linear regression.
- **Independence Assumption:** We discussed the assumption of independence in our model and its implications, noting that while it simplifies the problem, it may not hold true in all scenarios.
- **Model Selection:** The importance of evaluating our modeling assumptions in context was emphasized, especially concerning the choice of models and their appropriateness for the data at hand.